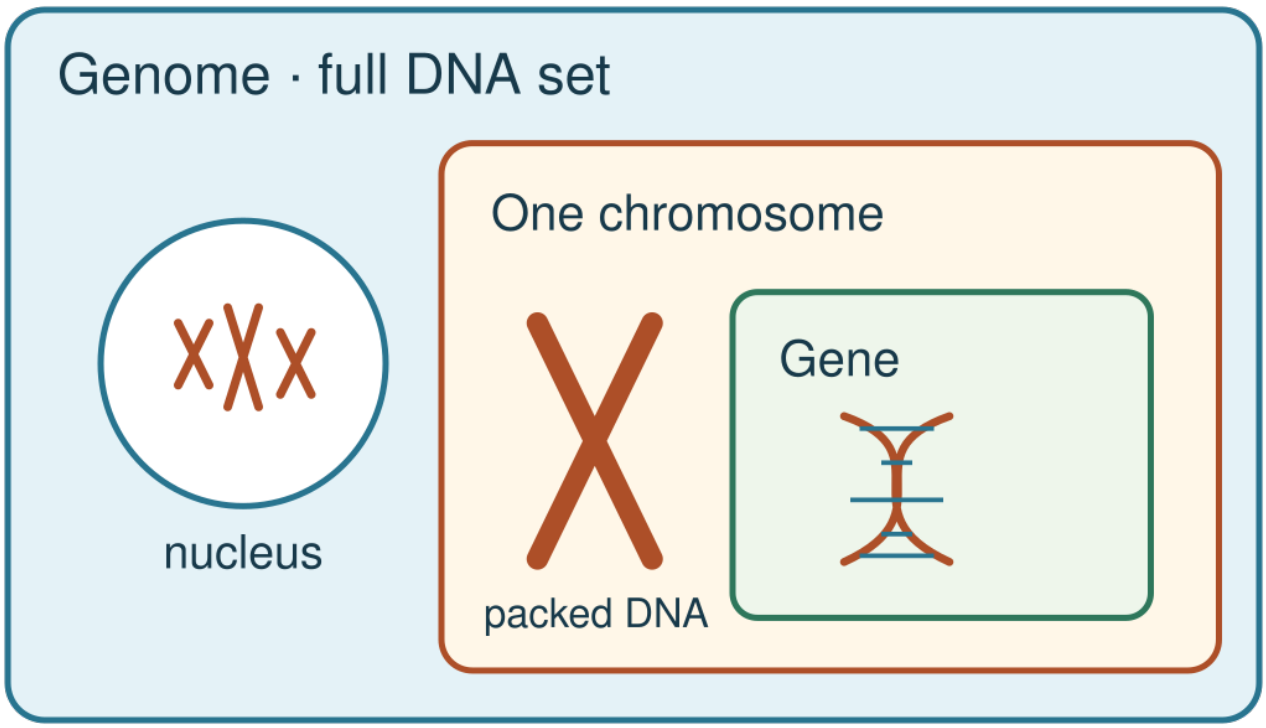
DNA, genes and chromosomes

Today I can… I can label DNA, a gene and a chromosome in a simple model. I can explain that a gene is a section of DNA with a protein instruction. I can distinguish one chromosome from the genome.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Knowledge organiser · model

I am learning to show how a gene is part of DNA and how DNA is packed into chromosomes.



The genome is the full DNA set; a chromosome carries part of it; a gene is a section of DNA on a chromosome. Nested view; not to scale.

Picture colours: blue genome frame, orange chromosome and DNA cord, green gene frame. Colours mark parts of the model; they are not real cell colours.

A gene is a DNA section with a protein instruction. One chromosome is not the whole genome.

Made from the web lesson SCI-LAUNCH-A2-W04 · version _____

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Knowledge organiser · words

Word	Meaning
DNA	the molecule that stores genetic instructions
gene	a section of DNA that codes for a specific protein
chromosome	a long DNA molecule packaged with proteins in a cell
genome	the entire DNA of an organism

Mark a short DNA section as a gene. Pack the DNA cord as a chromosome. The genome is the whole DNA set, including other DNA.

No model shows the real sizes or all of a real cell. You may use words in any language to plan your explanation.

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